

Wei-En Hsieh

+33 7 81 93 29 83 — ann20010929@gmail.com — Portfolio — GitHub — LinkedIn

Available full-time from July 2027 for 13–17 weeks

EDUCATION

Federal Institute of Technology Zurich (ETH Zürich)

Master of Science in Cyber Security, Minor in Data Management Systems

Sep 2024 - Expected Feb 2028

📍 Zurich, Switzerland

- Relevant Coursework: Advanced Systems Lab, Cloud Computing Architecture, Security Engineering

Federal Polytechnic School of Lausanne (EPFL)

Bachelor of Science in Computer Science

Sep 2021 - Jul 2024

📍 Lausanne, Switzerland

- Relevant Coursework: Algorithms, Operating Systems, Parallelism and Concurrency, System Programming

WORK EXPERIENCE

ETH Zürich — Information Security Group

Incoming Research Assistant Python

Oct 2026 – Feb 2027

📍 Zurich, Switzerland

- Selected to build an LLM-powered pipeline for automated Python instrumentation and compliance auditing.

Bouygues Telecom

Automation Project Manager — Software Engineering Python, Camunda, BPMN

Mar 2026 - Sep 2026

📍 Paris, France

- Automated an end-to-end Camunda workflow handling 30–50 mobile-network complaints/day, leaving only 2 manual exception paths.
- Orchestrated 4 APIs (15 calls/complaint) with retries, reusable components, and Gemini-based report analysis.
- Owned delivery from telecom-expert requirements to production, translating them into validated BPMN workflows.

WasteFlow

Full-Stack Developer TypeScript, Python, SQL

Jul 2023 - Jun 2024

📍 Lausanne, Switzerland

- Built a React/Node.js platform for 25+ recycling-factory machines, surfacing machine failures to operators.
- Partnered with 10+ domain experts to refine requirements and validate the platform against factory workflows.
- Implemented interactive dashboards to analyze machine performance and identify operational patterns.

EPFL

Teaching Assistant in Human-Computer Interaction (HCI), Linear Algebra, Mechanics

Feb 2024 - Jun 2024

📍 Lausanne, Switzerland

- Mentored 40+ HCI students building web and mobile UI/UX prototypes, reviewing functionality, usability, and implementation.
- Taught and supported students in Linear Algebra and Mechanics through exercises and problem solving.

RESEARCH & ENGINEERING PROJECTS

Reliable & Privacy-Preserving File Systems Python, JavaScript, SQL

Sep 2025 - Apr 2026

- Built GDPRFS, a FUSE file system enforcing runtime policies without modifying applications that access files.
- Delivered 0–80 ms enforcement overhead across 100–10,000 files, excluding LLM inference.
- Implemented page/row redaction on revocation and Lex mappings from FUSE events to GDPR concepts.
- Contributed the GDPRFS case study to research accepted at ACM CCS 2026 (Cycle B).

High-Performance Parallel Algorithm Optimization for Texture Synthesis C/C++, Python

Feb 2025 - Jun 2025

- Optimized the algorithm with AVX intrinsics on Linux, achieving 13× speedup and 7+ ops/cycle on the CPU.
- Automated Linux-based profiling and benchmarking with perf to detect bottleneck.

WanderPals: Full-Stack Android Platform for Group Travel Kotlin

Feb 2024 - Jun 2024

- Improved system reliability with almost 90% automated test coverage by implementing CI/CD pipelines.
- Built secure and scalable backend services with Firebase to ensure reliable group coordination across users.

LEADERSHIP & VOLUNTEERING

Paris Organising Committee for the 2024 Olympic and Paralympic Games

Volunteer, Sport Info & Administration

Jul 2024 - Sep 2024

📍 Paris, France

- Directed athlete operations for Marathon and Table Tennis, ensuring flawless execution before global audiences.
- Escorted medalists to the stage, ensuring precise, on-time podium ceremonies under global visibility.

SKILLS

Languages

French (bilingual), Mandarin (native), English (professional), German (A2)

Programming & Markup

Python, C, C++, Java, JavaScript, TypeScript, Kotlin, SQL, HTML, CSS, YAML

Frameworks & Technologies

Docker, Git, GitHub Actions, Figma, React.js, Node.js, Flask, MySQL, Bash, JSON

Operating Systems

Unix/Linux, Windows